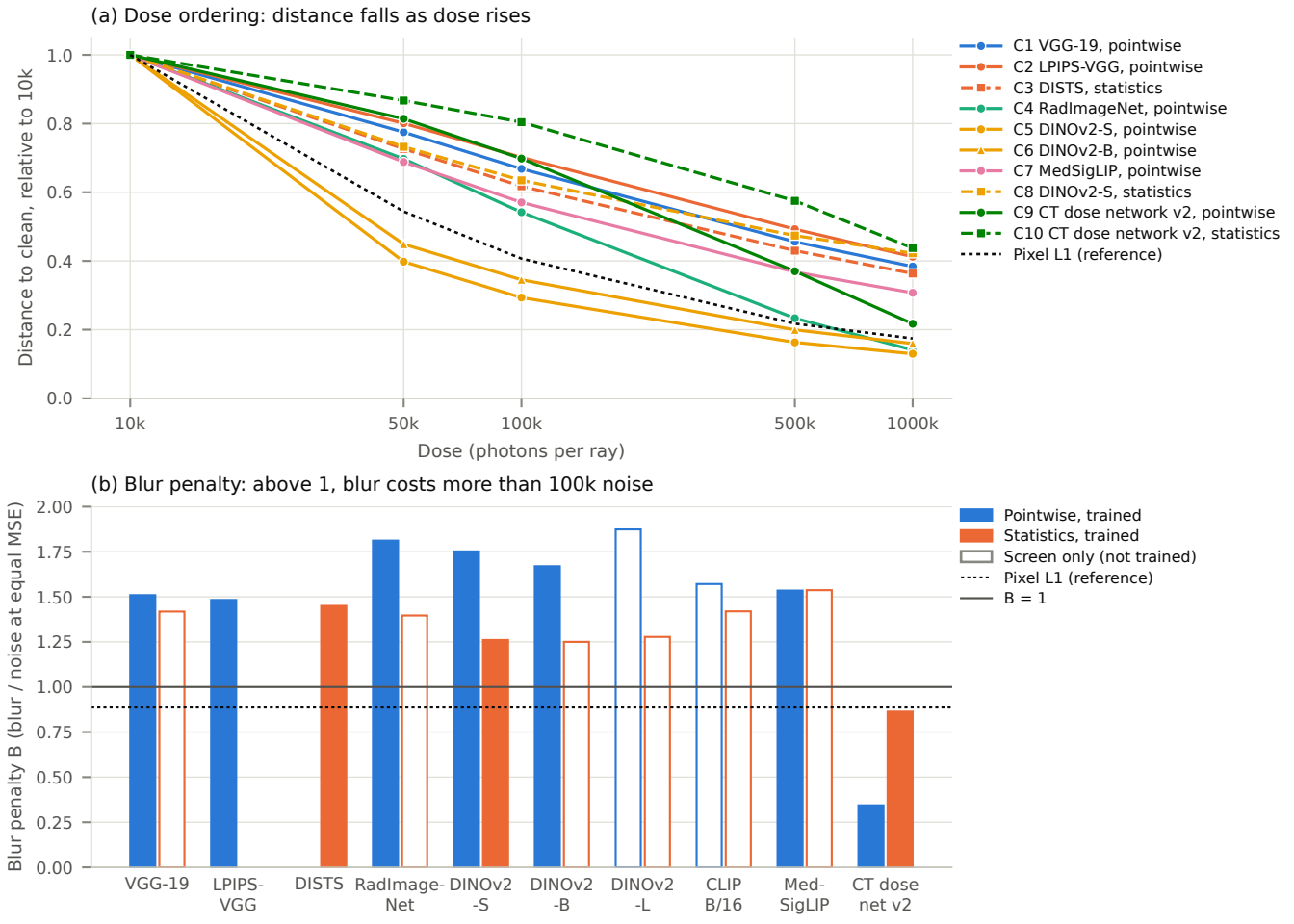
(a) Dose ordering: distance falls as dose rises
Distance to clean, relative to 10k
1.0
0.8
0.6
0.4
0.2
0.0
10k
50k
100k
500k
1000k
Dose (photons per ray)
C1 VGG-19, pointwise
C2 LPIPS-VGG, pointwise
C3 DISTS, statistics
C4 RadImageNet, pointwise
C5 DINOv2-S, pointwise
C6 DINOv2-B, pointwise
C7 MedSigLIP, pointwise
C8 DINOv2-S, statistics
C9 CT dose network v2, pointwise
C10 CT dose network v2, statistics
Pixel L1 (reference)
(b) Blur penalty: above 1, blur costs more than 100k noise
Blur penalty B (blur / noise at equal MSE)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
VGG-19
LPIPS-VGG
DISTS
RadImage-Net
DINOv2-S
DINOv2-B
DINOv2-L
CLIP B/16
Med-SigLIP
CT dose net v2
Pointwise, trained
Statistics, trained
Screen only (not trained)
Pixel L1 (reference)
B = 1